

Exhibition Guide — VIT Bhopal University

Exhibition Overview

Property	Value
Project	Neuromorphic Sleep Stage Scoring
Venue	VIT Bhopal University
Duration	3–5 minutes
Format	Live demonstration + poster
Final Model	Improved Student (99,477 parameters)

3–5 Minute Demo Script

Minute 1: Problem Statement (60 seconds)

Script:

"Sleep staging is clinically important but requires expert analysis of multi-channel polysomnography recordings. Manual scoring takes 2–4 hours per study and requires specialized training. We built an automated system to classify five sleep stages from physiological signals."

Show:

- Raw PSG waveform (EEG, EOG, EMG)
- Five sleep stages (Wake, N1, N2, N3, REM)
- Expert scoring timeline

Key points:

- Manual scoring is time-consuming
 - Expert training is required
 - Automation can assist clinicians
-

Minute 2: Signal Processing (60 seconds)

Script:

"Our pipeline processes raw PSG signals through bandpass filtering, notch filtering, and normalization. We apply artifact quality control and segment the signals into 30-second epochs."

Show:

- Before/after waveform comparison
- Filtering effects visualization
- Epoch segmentation

Key points:

- 0.5–35 Hz bandpass removes noise
- 50 Hz notch eliminates power-line interference
- Z-score normalization standardizes signals
- ~2% artifact flag rate

Minute 3: Model Architecture (60 seconds)

Script:

"The model uses a multi-resolution front end to capture patterns at different temporal scales. Depthwise-separable convolutions reduce computation while maintaining feature extraction. A compact Gabor block learns frequency-localized patterns. A two-layer GRU models temporal dependencies across five minutes of context."

Show:

- Architecture diagram
- Multi-resolution stem visualization
- Feature extraction flow
- GRU temporal modeling

Key points:

- Multi-scale feature capture
- Efficient depthwise-separable convolutions
- Learnable frequency filters
- 300-second temporal context

Minute 4: Live Inference (60 seconds)

Script:

"Let's run the model on a sample sequence. The model processes ten consecutive epochs and predicts the sleep stage for each."

Show:

- Load sample PSG sequence
- Run inference
- Display predictions with confidence
- Show hypnogram-like timeline

Demo commands:

```
# Run inference
python scripts/infer.py \
    --checkpoint artifacts/student_improved_best.pt \
    --input demo/sample_inputs/sample_epoch.npz
```

Expected output:

```
Predicted sleep stages:
Epoch  1: Wake    (confidence: 95.2%)
Epoch  2: N1     (confidence: 78.3%)
Epoch  3: N2     (confidence: 89.1%)
...
```

Minute 5: Results & Impact (60 seconds)

Script:

"Our final model achieves 87.5% test accuracy with Cohen's kappa of 0.763 across 15 subjects from Sleep-EDF Expanded. The model has only 99,477 parameters — small enough for edge deployment on microcontrollers. Most notably, we achieved 72% F1 on N1 sleep staging, up from 0% in the baseline, by expanding subject diversity. This demonstrates that efficient deep learning can automate sleep classification while remaining practical for resource-constrained devices."

Show:

- Official result block
- N1 improvement (0% → 72%)
- Parameter count comparison
- Latency measurement
- Edge deployment potential

Key result block:

87.5%
TEST ACCURACY

0.763
COHEN'S κ

0.720
N1 F1 SCORE

99,477
PARAMETERS

8.5 ms/batch
CPU LATENCY

Exhibition Poster Layout

Section A: Title

Neuromorphic Sleep Stage Scoring

A Compact Deep Learning Pipeline for Five-Stage Sleep Classification from PSG Signals

Section B: Problem

- Sleep staging requires expert analysis
- Manual scoring is time-consuming
- Automation can assist clinicians

Visual: Raw PSG waveform with stage annotations

Section C: Dataset

Sleep-EDF
EEG + EOG + EMG
11,128 processed epochs
Subject-level split

Section D: Signal Processing

Raw signal
↓
0.5-35 Hz bandpass
↓
50 Hz notch
↓
z-score
↓
artifact QC
↓
30 s epochs

Section E: Architecture

Multi-Resolution Stem
↓
Depthwise-Separable CNN
↓
Parametric Gabor FEB
↓
2-Layer GRU
↓
5-Class Output

Section F: Results

Large typography:

```
87.5%
TEST ACCURACY

0.763
COHEN'S  $\kappa$ 

0.720
N1 F1 SCORE

99,477
PARAMETERS
```

Section G: LoRA Adaptation

Parameter-Efficient Adaptation:

```
LoRA r=8: 552 trainable params (0.55%)
  →  $\kappa$  = 0.8092 (94.2% of full fine-tuning)
  → Merge diff: 0.00e+00
  → Adapter reload: 0.00e+00
```

Key message: "The same 99K-parameter backbone can be adapted to new subjects using only 552 trainable parameters — less than 1% of the full model."

Section G: Deployment

```
Compact model
  ↓
Low memory footprint
  ↓
Fast CPU inference
  ↓
Edge / MCU deployment
```

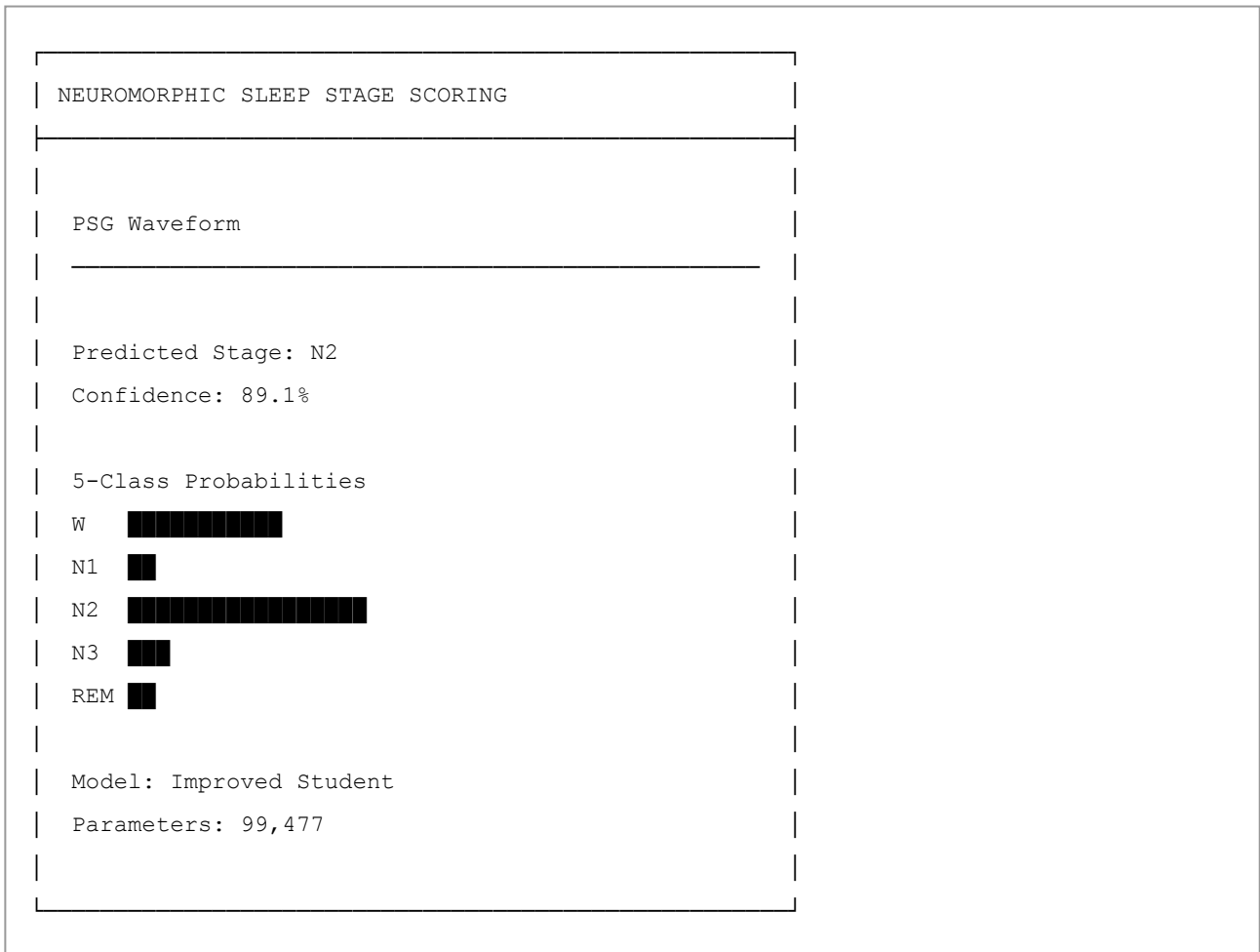
Section H: Team

Member	Role
Param Kaushik	Dataset & Data Governance
Suha Vora	Signal Preprocessing
Shailendra Bhatt	Exploratory Data Analysis

Member	Role
Shamique Khan	Model Development & Training
Aasir Jaffer Lone	Evaluation & Performance

Live Demo Dashboard

Streamlit Layout



Demo Buttons

- [Load Sample] — Load a sample PSG sequence
 - [Run Inference] — Run model prediction
 - [Show Signal] — Display multi-channel waveforms
 - [Show Probabilities] — Display class probabilities
 - [Show Hypnogram] — Display predicted timeline
-

Q&A Preparation

Technical Questions

Q: Why use a 300-second context window? A: Sleep stages are temporally dependent. 10 consecutive 30-second epochs provide 5 minutes of context, allowing the GRU to model stage transitions.

Q: Why not classify each epoch independently? A: Isolated epochs can be ambiguous, especially around transitions ($W \rightarrow N1$, $N1 \rightarrow N2$). Sequence modeling provides contextual information.

Q: Why use EEG, EOG, and EMG together? A: Each channel provides complementary information: EEG for brain activity, EOG for eye movements, EMG for muscle activity.

Q: Why depthwise-separable convolutions? A: They reduce computational cost by ~77% compared to standard convolutions while maintaining feature extraction capability.

Q: Why is accuracy not enough? A: Class imbalance means a model can have high accuracy while performing poorly on minority stages. Cohen's kappa and F1 are more informative.

Deployment Questions

Q: Can this run on a microcontroller? A: Yes. The model has 99,477 parameters (~400 KB at FP32) and 8.5 ms latency, suitable for ARM Cortex-M7 class devices.

Q: What is the memory footprint? A: ~400 KB for weights, plus ~120 KB for input buffer. Total < 1 MB, fitting on most microcontrollers.

Q: Can it run in real-time? A: Yes. At 8.5 ms per batch with 30-second epochs, the model has significant headroom for real-time inference.

Limitation Questions

Q: What are the main limitations? A: N1 and REM classification are challenging due to class imbalance and subtle signal differences. The model is trained on healthy subjects only.

Q: Is this clinically validated? A: No. This is a research prototype. Clinical validation would require larger, more diverse datasets and regulatory approval.

Checklist

Before Exhibition

- ☐ Live demo working on exhibition machine
- ☐ Streamlit dashboard tested
- ☐ Sample data files available
- ☐ Poster printed and mounted

- ☐ Backup slides prepared
- ☐ Team roles assigned for Q&A

During Exhibition

- ☐ Start with problem statement
- ☐ Show signal processing pipeline
- ☐ Demonstrate live inference
- ☐ Present official result
- ☐ Answer questions confidently
- ☐ Acknowledge limitations honestly

After Exhibition

- ☐ Collect feedback
- ☐ Document any issues
- ☐ Update project based on feedback
- ☐ Share results with team

